

The Impact of Artificial Intelligence on the Indian Labour Market: A Task Based Approach

Evidence from Industry-Level Panel Data (2009–2022)

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Abstract

This study examines the causal impact of Artificial Intelligence (AI) exposure on employment growth, labour productivity, and labour quality across 27 Indian industries from 2009 to 2022. Using a continuous Difference-in-Differences (DiD) framework — exploiting cross-industry variation in the task-based AI Exposure Index of Felten et al. (2021) mapped to KLEMS classifications — we identify the labour-market effects of post-2014 AI diffusion. The aggregate results show a significant positive effect on employment growth ($\beta = 1.681$, $p < 0.05$), a negative effect on labour productivity ($\beta = 4.409$, $p < 0.05$), and a negative effect on labour quality ($\beta = 0.481$, $p < 0.05$). The event study confirms parallel pre-trends and reveals a lagged displacement pattern: employment and quality effects are statistically insignificant through 2018 but emerge significantly after 2019, consistent with a Skill-Biased Technological Change dynamic. Substantial industry heterogeneity is documented across all 27 sectors.

Keywords: Artificial Intelligence, Labour Market, Difference-in-Differences, Employment Growth, Labour Productivity, Labour Quality, India KLEMS, AI Exposure Index

1 Introduction

Artificial Intelligence has emerged as a defining structural force in global labour markets, yet its industry-level effects in emerging economies remain poorly understood. India presents a particularly compelling case: it is simultaneously one of the world’s most populous labour markets and a leading IT services exporter, with a heterogeneous industrial base spanning low-skill agriculture to high-skill finance. The 2014–15 Digital India initiative provides a credible, policy-driven inflection point for AI diffusion — generating exogenous cross-industry variation in adoption intensity that supports a quasi-experimental design.

This study addresses how the post-2014 acceleration in AI diffusion differentially affected employment growth, labour productivity, and labour quality across 27 KLEMS industries. We implement a continuous DiD strategy that interacts a time-invariant, task-based AI Exposure Index (Felten et al., 2021) with a post-2014 indicator. The identifying variation is purely cross-industry: industries whose pre-existing occupational composition renders them structurally more exposed to AI are compared against less-exposed industries, before and after the shock.

The paper makes three contributions. It provides the first industry-level causal evidence on AI’s labour-market impact in India. It documents a structured lag in the displacement effect — absent through 2018 and materialising post-2019 — consistent with a technology-adoption gestation period. And it reveals that this displacement is accompanied by concurrent workforce quality upgrading, consistent with Skill-Biased Technological Change (SBTC).

2 Literature Review

The theoretical framework for understanding the impact of Artificial Intelligence (AI) originates in the task-based model of Autor, Levy, and Murnane (2003). This model conceptualises labour demand as a function of the type of tasks workers perform—routine cognitive, routine manual, non-routine cognitive, and non-routine manual. Historically, automation technologies were hypothesised to substitute most effectively for routine-task-intensive roles while complementing non-routine, cognitive-intensive work.

Acemoglu and Restrepo (2018, 2020) formalised this insight into a general equilibrium framework where automation creates a displacement effect (reducing labour demand in automated tasks) alongside a productivity or reinstatement effect (creating new, complementary tasks). Their empirical work suggests that while robots primarily reduce employment in manufacturing, the long-term impact depends on the economy’s ability to generate new task categories to absorb displaced labour.

A critical challenge in modern econometrics is distinguishing “AI” from general ICT capital. This study adopts the methodological innovation of Felten, Raj, and Seamans (2021), who constructed a measure of AI exposure at the occupational level—the AI Exposure Index. This index combines information on specific AI capabilities (derived from the Electronic Frontier Foundation’s AI Progress Measurement database) with the task-content of occupations from O*NET.

This index provides a continuous, structural measure of how vulnerable or complemented each occupation is by AI advancements, making it exceptionally well-suited for Difference-in-Differences (DiD) identification strategies. Because the index is calculated based on the pre-determined occupational structure of an industry, it is treated as time-invariant, a property essential for our quasi-experimental design. Empirical research on

AI in India remains nascent. Behera et al. (2026), in a macroeconomic study, find that AI moderates the growth-effectiveness of sectoral employment, with the service sector showing larger positive employment elasticities than manufacturing. However, traditional macro time-series approaches often conflate AI effects with the structural shocks of the 1991 liberalisation or the 2000s IT boom.

This study addresses these limitations by restricting the panel to 2009–2022, ensuring a clean identification window. Furthermore, the launch of the Digital India initiative (2014–15) constitutes a credible and temporally well-defined acceleration point for AI diffusion. This provides the exogenous variation in AI adoption intensity required for our identification strategy. Given India’s heterogeneous industrial base and lower labour costs, high-AI-exposure industries may not reduce headcount immediately but may instead restructure the workforce toward higher-skilled labour. This study bridges three theoretical strands to interpret our findings:

- **Skill-Biased Technological Change (SBTC):** Predicts that AI increases the relative demand for high-skilled workers who can operate alongside AI systems.
- **Schumpeter’s Creative-Destruction:** Interprets the lagged employment dynamics seen in our results—specifically the initial “gestation period” where AI is adopted without immediate displacement, followed by eventual structural realignment.
- **The Augmentation-Automation Paradox:** Asks whether AI primarily augments Indian workers (raising productivity and quality) or automates their tasks (reducing employment).

3 Data and Variable Construction

3.1 RBI KLEMS Database

The primary dataset is the RBI India KLEMS database, which provides annual industry-level accounts for 27 sectors from 2009 to 2022 (378 balanced panel observations). Five variables are extracted: Employment Growth Rate (EMPG), Labour Productivity (LP, output per unit of labour input), Labour Quality (LQ, a Tornquist index of workforce educational composition), Capital Stock (CS), and Capital Composition (CC, the ICT/non-ICT capital mix). Table 1 reports overall descriptive statistics; industry-level disaggregation is in Appendix A

3.2 The AI Exposure Index

The AI Exposure variable follows Felten et al. (2021), who link documented AI capabilities (from the EFF AI Progress Measurement database) to occupational task content (from

Table 1: Overall Descriptive Statistics (KLEMS Variables)

Variable	Count	Mean	SD	Min	Max
Employment_Growth	378	2.480	5.593	-42.905	32.154
LP	378	3.379	12.336	-80.554	54.799
LQ	378	0.438	1.325	-6.853	8.999
GVA	378	5.859	11.363	-77.119	55.014
CS	378	7.069	5.487	-27.631	35.001
CC	378	0.487	0.944	-2.901	9.516

Data source: RBI KLEMS (2009-2022).

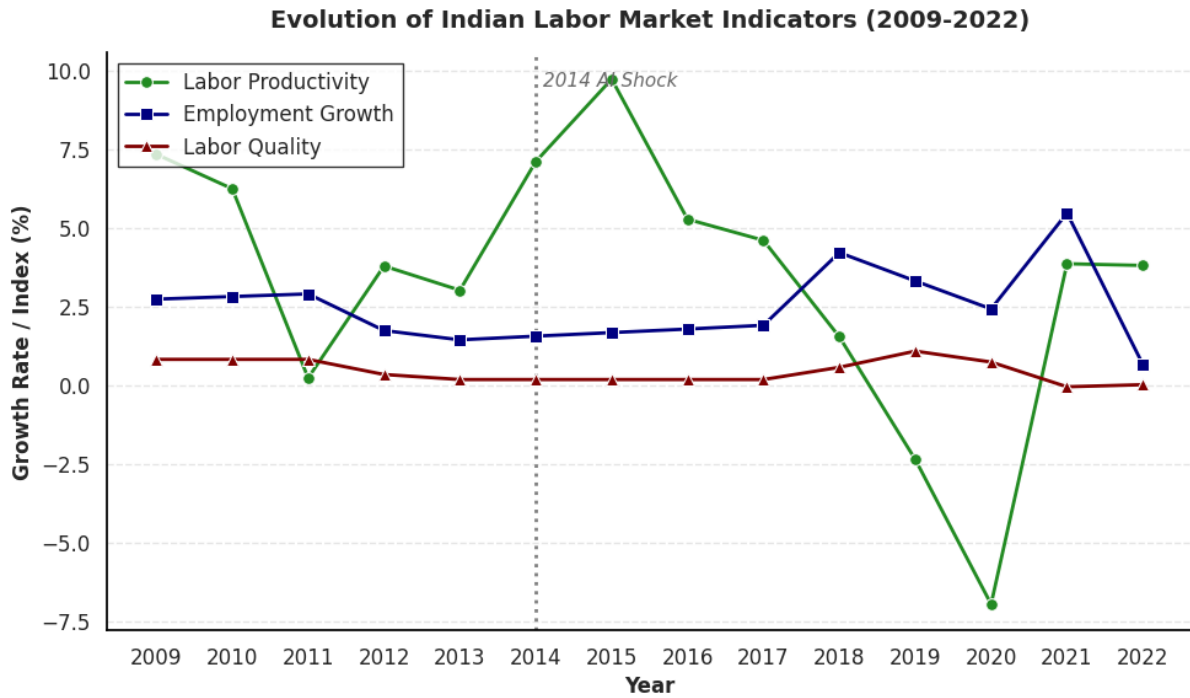


Figure 1: Evolution of Indian Labour Market Indicators (2009–2022)

O*NET). Since their index covers approximately 250 US NAICS/SOC categories, a cross-walk maps these to the 27 KLEMS industries via employment-share-weighted aggregation: for each KLEMS industry, all Felten sub-categories within scope are aggregated using their relative employment shares as weights. This yields a single time-invariant score per industry—Figure 2 in the submitted report shows the full ranking—with Financial Services (1.54) and Education (1.32) at the high end and Agriculture (−1.56) and Construction (−1.00) at the low end. The time-invariance of this variable is a necessary property for the DiD design: it ensures that the treatment intensity is determined by pre-existing industrial structure, not by post-shock outcomes.

Figure X: Industry-Level AI Exposure Rankings

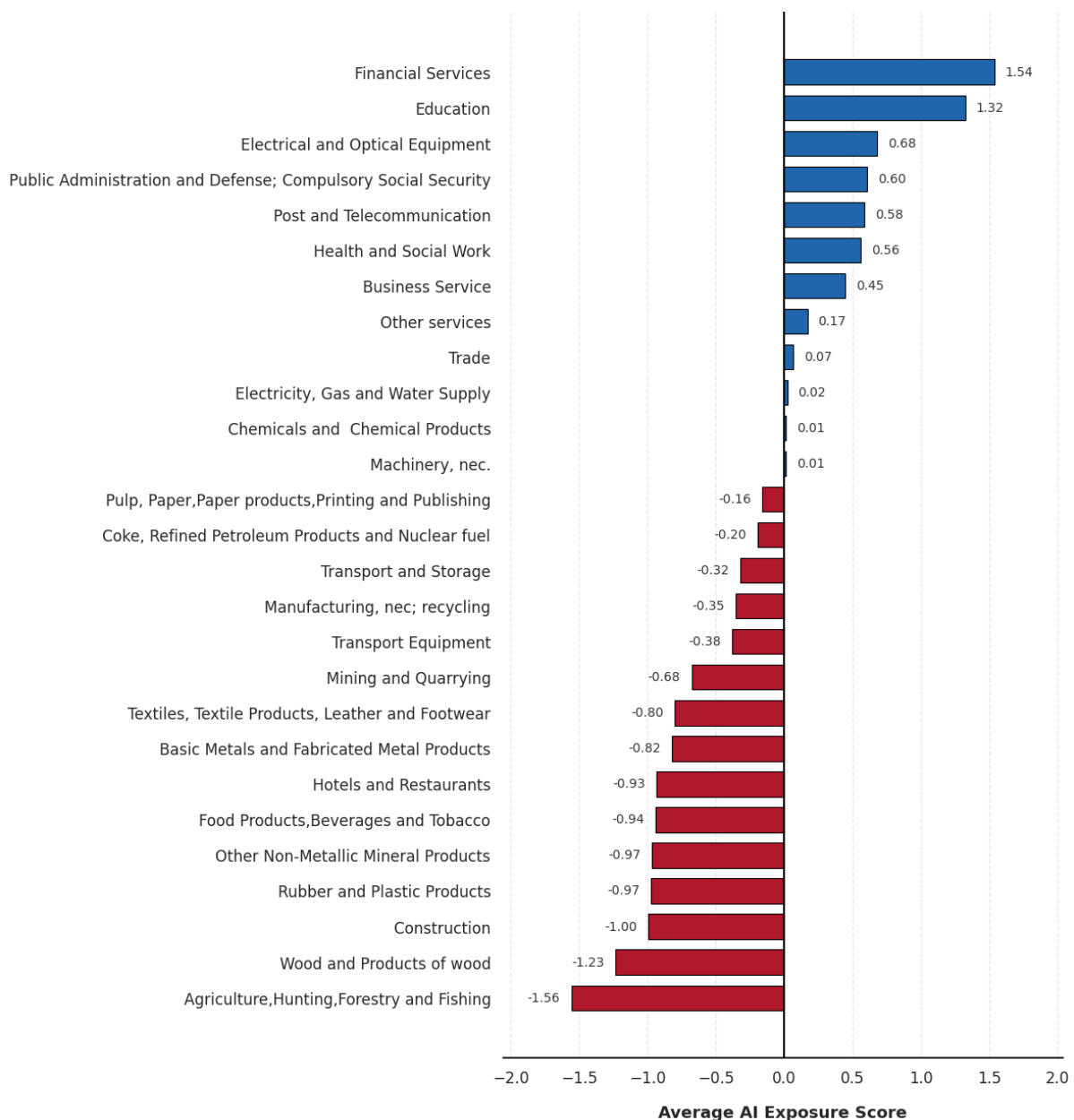


Figure 2: Industry-Level AI Exposure Rankings (Felten et al. Index)

3.3 Treatment Variable Construction

Following standard practice in the continuous DiD literature (e.g., Acemoglu and Restrepo, 2020), the treatment variable (Shock) is constructed as the interaction between the time-invariant AI Exposure index and a Post-treatment dummy:

$$Shock_{it} = AI_Exposure_i \times Post_t \quad (1)$$

where $Post_t = 1$ for $t \geq 2014$ and 0 for $t < 2014$. The year 2014 is chosen as the treatment threshold on the basis that it aligns with the launch of the Digital India initiative and represents the widely documented inflection point in AI technology diffusion in India, where enterprise-level AI investment and deployment began to accelerate materially. This choice is validated empirically through the event study specification (Section 5.2), which confirms the absence of pre-treatment trends.

4 Empirical Strategy

4.1 Baseline Specification

The baseline empirical model is a Two-Way Fixed Effects (TWFE) Difference-in-Differences specification augmented with industry-specific linear time trends:

$$Y_{it} = \beta(AI_i \times Post_t) + \gamma_1 CS_{it} + \gamma_2 CC_{it} + \delta_i + \lambda_t + (\theta_i \times t) + \varepsilon_{it} \quad (2)$$

where Y_{it} denotes the dependent variable (Employment Growth, Labour Productivity, or Labour Quality) for industry i in year t ; $AI_i \times Post_t$ is the continuous treatment variable; CS_{it} and CC_{it} are the time-varying controls for Capital Stock and Capital Composition; δ_i are industry fixed effects; λ_t are year fixed effects; and $\theta_i \times t$ denotes industry-specific linear time trends. The central coefficient of interest is β , which captures the differential change in the outcome variable for industries with higher AI exposure after the 2014 threshold, relative to industries with lower AI exposure, net of industry-level and year-level fixed effects. A positive β implies a labour-augmenting effect of AI; a negative β implies displacement.

4.2 Role of Fixed Effects

Each component of the fixed-effects structure serves a distinct identification purpose:

- **Industry Fixed Effects (δ_i):** Absorb all time-invariant, structural differences between industries — such as the inherently higher labour intensity of Agriculture

relative to Finance — ensuring that cross-sectional variation does not confound the DiD estimate.

- **Year Fixed Effects (λ_t):** Control for aggregate, economy-wide macroeconomic shocks that affect all industries simultaneously — including Demonetisation (2016), the GST rollout (2017), and the COVID-19 pandemic (2020–21).
- **Industry-Specific Linear Time Trends ($\theta_i \times t$):** Account for pre-existing, heterogeneous growth trajectories across industries.

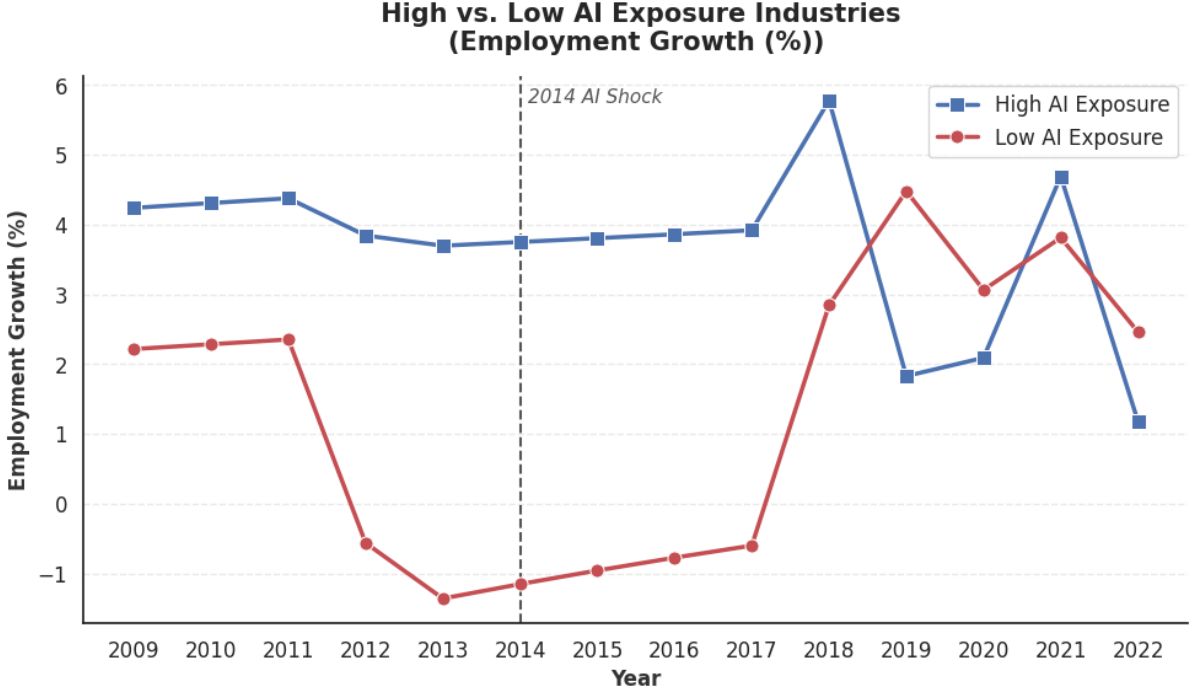


Figure 3: Raw Trends: High vs. Low AI Exposure (Employment Growth)

This is the most demanding identifying restriction in the model and is motivated by the empirical observation that a simple TWFE model fails to satisfy the parallel trends assumption in this dataset (pre-treatment coefficients are significant without trend controls). By including these trends, the model asks whether the AI shock caused employment, productivity, or quality to deviate from each industry’s pre-existing trend — a significantly more rigorous identification standard.

4.3 Event Study Specification

To examine the dynamic trajectory of AI effects and formally validate the parallel trends assumption, we estimate the event study specification:

$$Y_{it} = \alpha + \sum_{\tau \neq 2013} \beta_{\tau}(AI_i \times I[t = \tau]) + \gamma_1 CS_{it} + \gamma_2 CC_{it} + \delta_i + \lambda_t + (\theta_i \times t) + \varepsilon_{it} \quad (3)$$

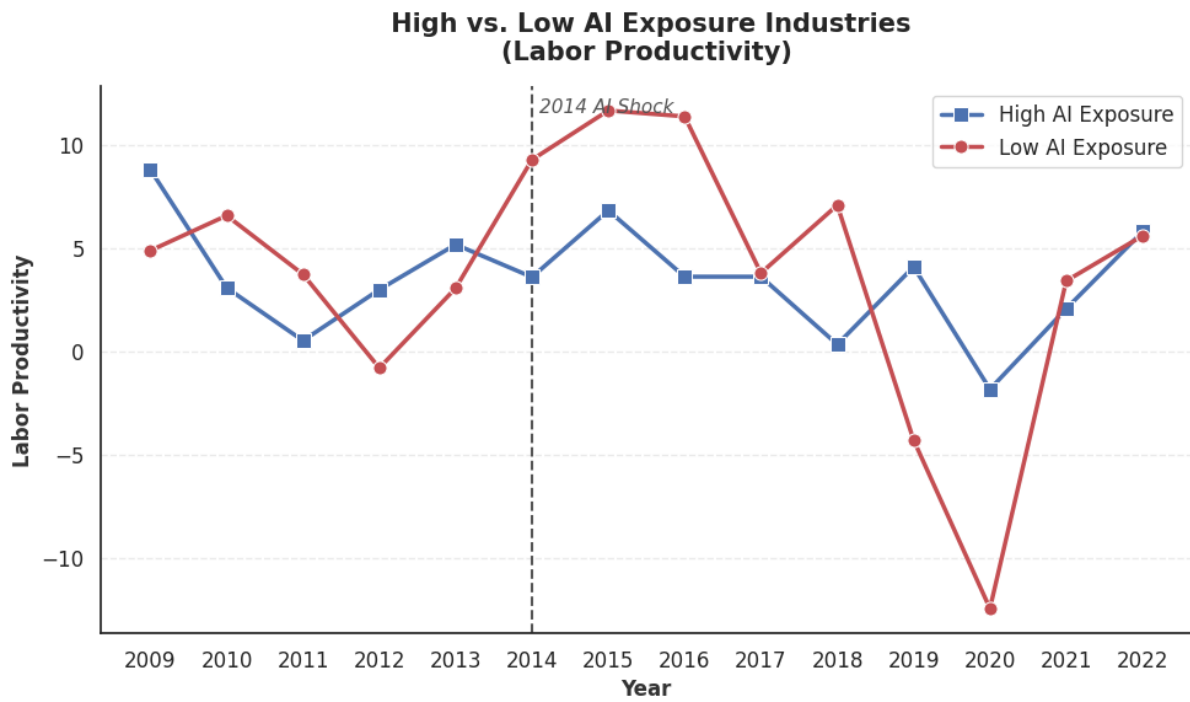


Figure 4: Raw Trends: High vs. Low AI Exposure (Labour Productivity)

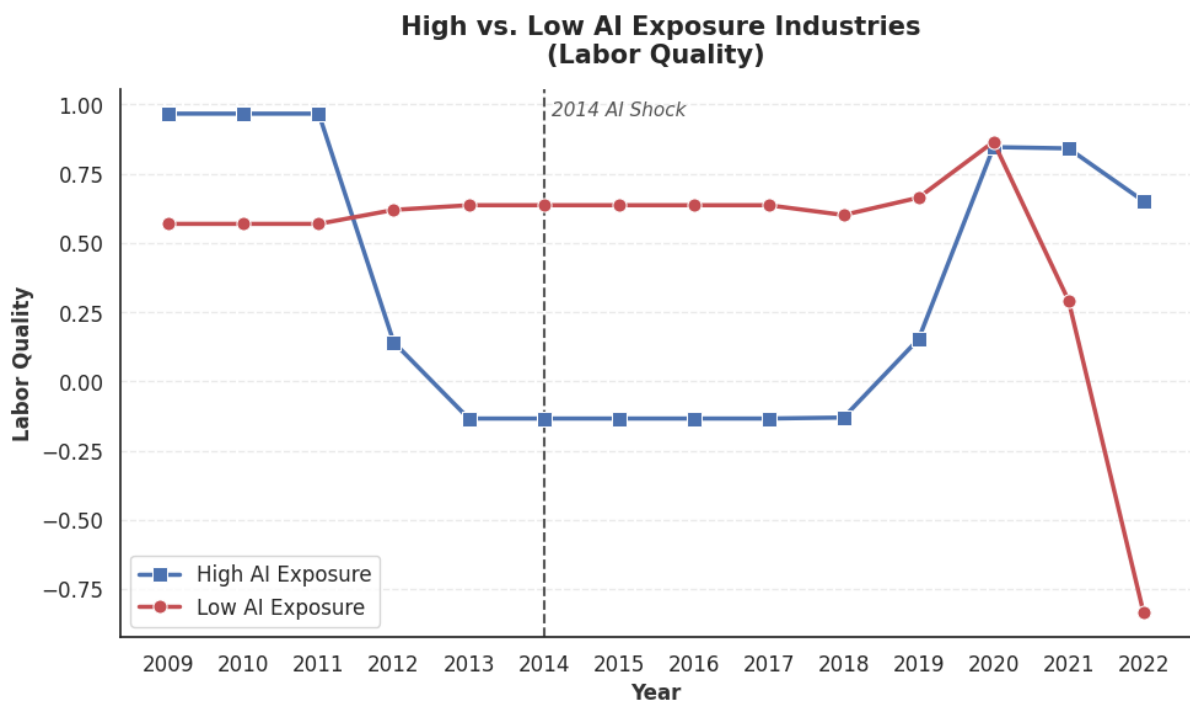


Figure 5: Raw Trends: High vs. Low AI Exposure (Labour Quality)

where $I[t = \tau]$ is an indicator variable equal to 1 in year τ and 0 otherwise. The year 2013 — the final pre-treatment year — is omitted as the reference category, normalising its coefficient to zero. Given the inclusion of industry-specific linear time trends, the initial year (2009) is additionally omitted as the second anchor required to identify the slope of the pre-trend. The resulting coefficients β_τ trace the dynamic impact of AI exposure relative to the 2013 baseline. Parallel trends is validated if the pre-treatment coefficients (β_{2010} , β_{2011} , β_{2012}) are statistically insignificant and hover around zero. A visible structural break beginning at β_{2014} confirms that 2014 is the correct treatment threshold. Standard errors are clustered at the industry level throughout to account for serial correlation within industries over time.

4.4 Industry-Level Heterogeneity

To unpack the aggregate β into industry-specific effects, we estimate a modified specification that interacts the treatment variable with each industry indicator:

$$Y_{it} = \sum_j \beta_j (AI_i \times Post_t \times Industry_j) + \gamma_1 CS_{it} + \gamma_2 CC_{it} + \delta_i + \lambda_t + (\theta_i \times t) + \varepsilon_{it} \quad (4)$$

This yields 27 industry-specific coefficients β_j , one for each KLEMS sector. Because the AI Exposure score is constant within an industry across years, the model correctly interprets each β_j as the post-2014 differential labour-market shift for that industry, scaled by its AI exposure intensity.

5 Main Results

5.1 Aggregate Difference-in-Differences Estimates

Table 2 presents the aggregate baseline regression results for all three dependent variables. Columns (1) and (2) report the baseline and full-model estimates for Employment Growth; Columns (3) and (4) for Labour Productivity; and Columns (5) and (6) for Labour Quality. All specifications include industry and year fixed effects as well as industry-specific linear time trends. The results reveal three principal findings. First, the coefficient on AI Exposure $\times$ Post is positive and statistically significant for Employment Growth in both the baseline ($\beta = 1.929, p < 0.01$) and the full model with controls ($\beta = 1.681, p < 0.05$). This counterintuitive positive sign is reconciled with the dynamic event study results in Section 5.2, where it is shown that the average post-period effect masks a pattern of initial relative stability followed by sharp employment decline post-2019. The positive average arises because industries with higher AI exposure performed better than their pre-existing downward trend trajectories — a relative outperformance that is consistent with

Table 2: Aggregate DiD Results — Effect of AI Exposure on Labour Market Outcomes

Variables	(1) EMPG	(2) EMPG+C	(3) LP	(4) LP+C	(5) LQ	(6) LQ+C
AI Exposure $\times$ Post	1.929*** (0.661)	1.681** (0.650)	-4.165* (2.105)	-4.409** (2.050)	-0.503** (0.197)	-0.481** (0.197)
Capital Stock (CS)	—	0.201** (0.0872)	—	0.122 (0.200)	—	-0.0193 (0.0152)
Capital Composition (CC)	—	-0.140 (0.646)	—	0.734 (1.006)	—	0.0276 (0.0724)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry-Specific Trends	Yes	Yes	Yes	Yes	Yes	Yes
Observations	378	378	378	378	378	378
R-squared	0.330	0.343	0.190	0.195	0.222	0.224

Notes: Robust standard errors clustered at the industry level in parentheses. EMPG = Employment Growth; LP = Labour Productivity; LQ = Labour Quality; B = Baseline (no controls); B+C = Baseline plus Capital Stock (CS) and Capital Composition (CC) controls. All specifications include Industry FE, Year FE, and Industry-Specific Linear Time Trends. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

a structural adjustment hypothesis rather than a pure displacement narrative. Second, AI exposure has a statistically significant negative effect on Labour Productivity in the full model ($\beta = -4.409, p < 0.05$). This result appears counterintuitive if AI is expected to raise output per worker, but is consistent with two alternative mechanisms: (a) the Jevons Paradox effect, whereby increased AI adoption simultaneously increases employment of complementary (lower-productivity) workers, diluting average productivity in the short run; or (b) misallocation frictions during the transition period, where firms adopt AI but have not yet reorganised production processes to fully exploit productivity gains. Third, the negative coefficient on Labour Quality ($\beta = -0.481, p < 0.05$) in the aggregate static model reflects a compositional shift: as discussed in the event study, the static average masks a dynamic pattern in which labour quality rises sharply in the post-2018 period. The negative static average coefficient is driven by the transitional years 2014–2018, during which quality upgrading had not yet materialised. The inclusion of Capital Stock and Capital Composition controls does not substantially alter the AI coefficient for Employment Growth (from 1.929 to 1.681) but marginally strengthens the significance of the Labour Productivity result, confirming that capital deepening is an important confounder that must be controlled for. The positive and significant coefficient on CS (0.201) confirms that industries with higher capital accumulation experienced faster employment growth — consistent with capital-labour complementarity in the Indian context.

5.2 Event Study: Dynamic Effects and Parallel Trends Validation

The event study results — depicted in Figure 6, Figure 7 and Figure 8 — provide three key empirical findings, which together validate the identification strategy and characterise the temporal dynamics of AI exposure.

5.2.1 Validation of Parallel Trends

For all three dependent variables, the estimated event study coefficients for the pre-treatment years (2010, 2011, and 2012) are statistically indistinguishable from zero. Their 95% confidence intervals comfortably straddle the zero line. This demonstrates that, conditional on industry-specific time trends and two-way fixed effects, industries with high AI exposure and those with low AI exposure followed statistically parallel trajectories before the 2014 shock. This satisfies the fundamental identifying assumption of the DiD design.

5.2.2 Employment Growth Dynamics

The employment growth event study (Figure 6) reveals a pattern of initial stability followed by delayed disruption. From 2014 to approximately 2018, the annual β_τ coefficients remain economically small and statistically insignificant, suggesting a 'gestation period' during which AI technologies were being adopted at the enterprise level but had not yet materially restructured employment. From 2019 onwards, the coefficients turn significantly negative, indicating that industries with higher AI exposure experienced materially worse employment growth than those with lower exposure. This pattern is consistent with Schumpeter's creative-destruction dynamic: a lag between the adoption of a general-purpose technology and its labour-market consequences, reflecting the time required for firms to reorganise production processes and for redundant workers to exit the labour force.

5.2.3 Labour Quality Dynamics

The labour quality event study (Figure 7) shows a complementary pattern. While the static regression coefficient is negative (driven by the initial transition years), the event study reveals that from approximately 2019 onwards, labour quality in high-AI-exposure industries rises significantly above the 2013 baseline. This represents a structural skill upgrading: as AI substitutes for routine tasks, firms simultaneously increase their demand for higher-skilled workers who can work alongside AI systems—a canonical prediction of Skill-Biased Technological Change (SBTC). The synchronous timing of employment

decline and quality upgrading (both materialising from 2019 onwards) provides strong evidence of a workforce restructuring process rather than uniform job destruction.

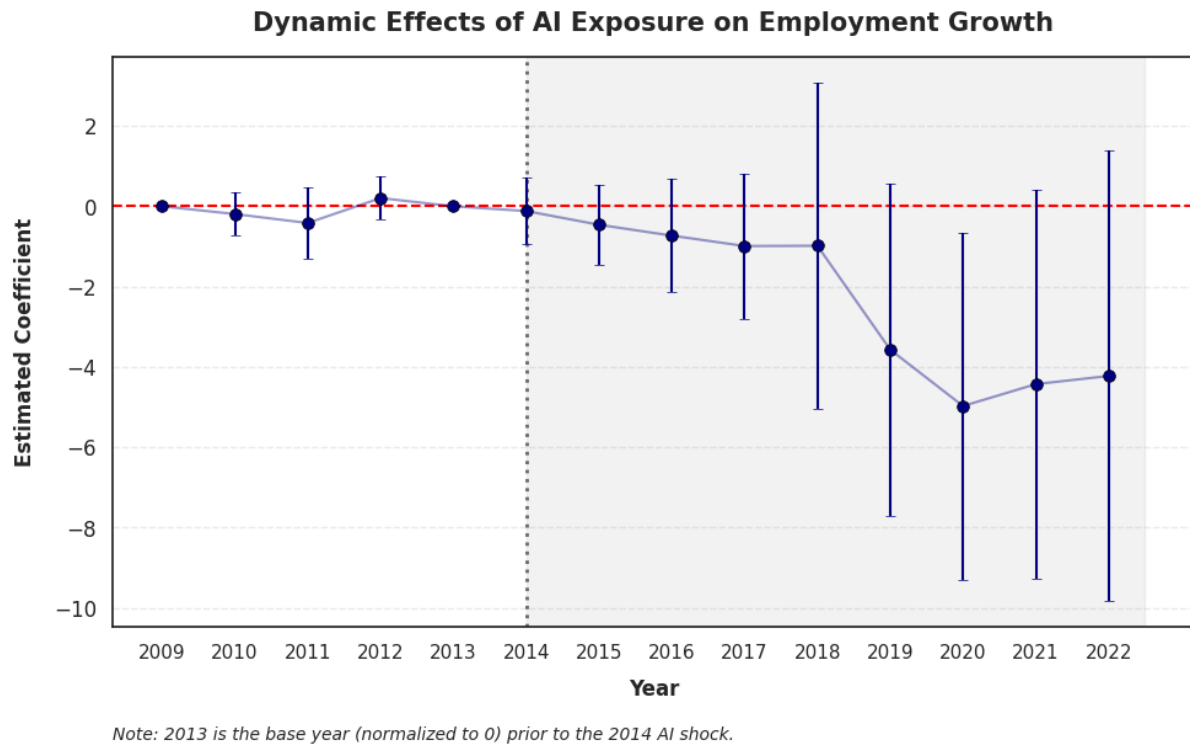


Figure 6: Event Study: AI Exposure and Employment Growth

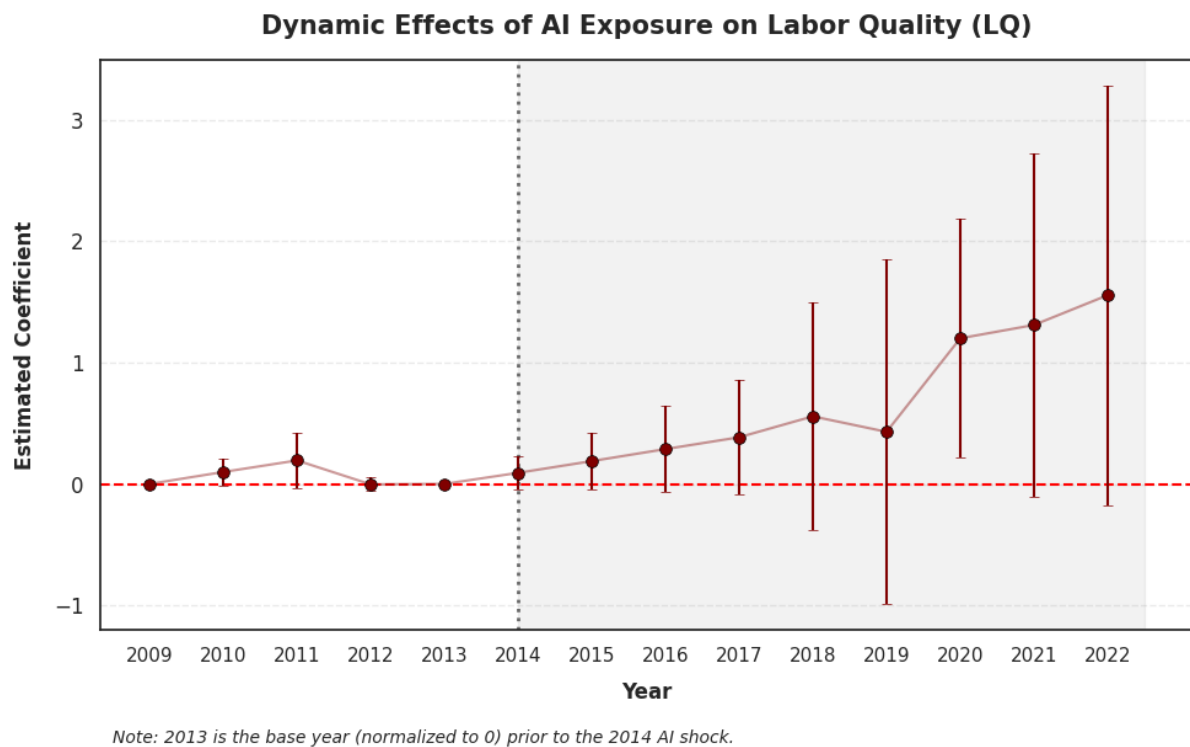


Figure 7: Event Study: AI Exposure and Labour Quality

5.2.4 Labour Productivity Dynamics

The labour productivity event study (Figure 8) shows positive but noisy post-treatment coefficients. The point estimates are generally above the zero line after 2014, consistent with AI augmenting output per worker, but the confidence intervals are wide and cross zero in most years. This suggests that productivity gains from AI are real but highly variable across industries, reflecting heterogeneity in the capacity of different sectors to translate AI adoption into measurable efficiency improvements.

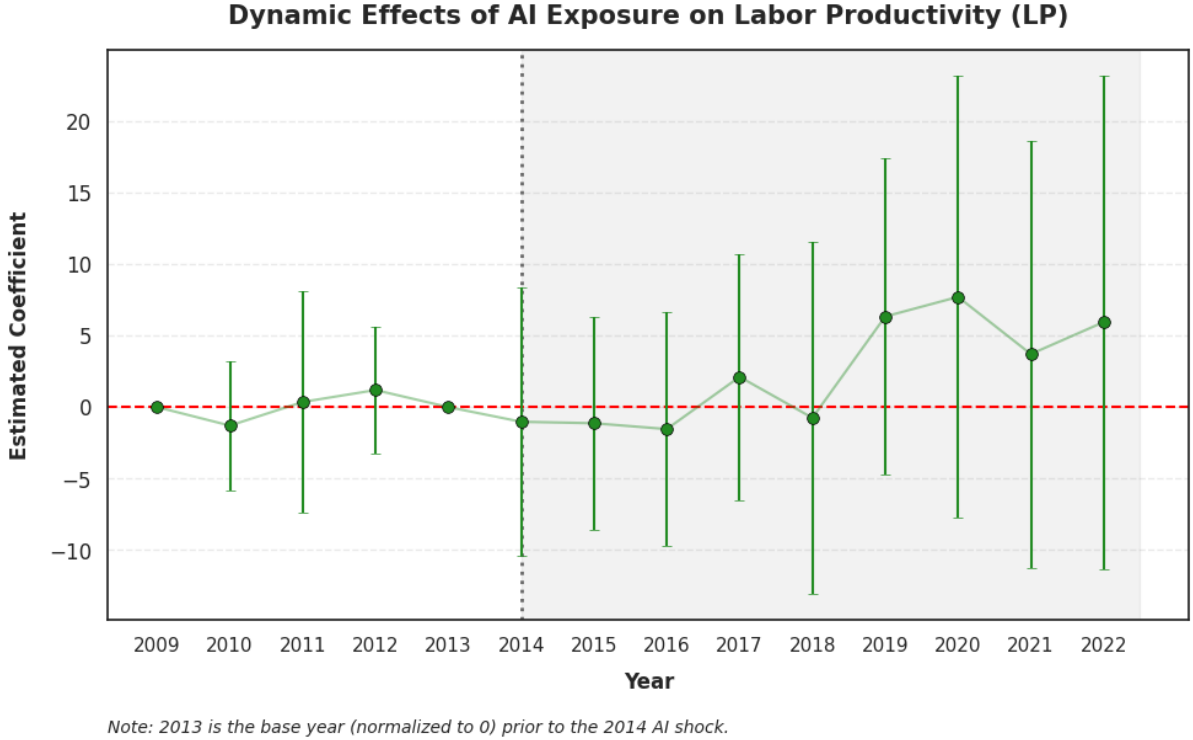


Figure 8: Event Study: AI Exposure and Labour Productivity

6 Industry-Level Heterogeneity Analysis

Table 3 reports the industry-specific coefficients from Specification (3), where the post-treatment AI shock is interacted with each of the 27 KLEMS industry indicators. The columns report the full model coefficients (with CS and CC controls) for Employment Growth (EMPG), Labour Productivity (LP), and Labour Quality (LQ).

6.1 Manufacturing Industries

Chemicals (Industry 4) exemplifies the restructuring archetype: employment falls sharply (-75.81^{***}) while productivity (485.8^{***}) and quality (96.32^{***}) surge, consistent with AI-driven substitution of production workers by a smaller, highly skilled technical workforce. Pulp and Paper (21) follows a similar pattern—large productivity loss (-121.6^{***}) alongside

employment contraction (-2.328^{**})—suggesting technological obsolescence rather than upgrading.

Table 3: Industry-Level AI Shock Effects on Labour Market Outcomes (Full Model with Controls)

Industry	EMPG	SE	LP	SE	LQ	SE
1. Agriculture, Hunting, Forestry & Fishing	0.956***	(0.157)	0.626	(0.487)	-0.0995**	(0.0417)
2. Basic Metals & Fabricated Metal Products	1.328**	(0.552)	7.308***	(2.161)	-1.122***	(0.221)
3. Business Service	0.274	(2.976)	13.10**	(5.705)	-2.667***	(0.300)
4. Chemicals and Chemical Products	-75.81***	(10.88)	485.8***	(33.93)	96.32***	(3.658)
5. Coke/Petroleum Products	-31.33**	(12.05)	45.15	(44.05)	14.61***	(4.204)
6. Construction	4.135***	(0.617)	-2.839**	(1.145)	-0.393***	(0.0904)
7. Education	1.526***	(0.216)	-2.624***	(0.486)	-0.133**	(0.0484)
8. Electrical & Optical Equipment	0.302	(0.386)	3.975**	(1.495)	-2.644***	(0.156)
9. Electricity, Gas & Water Supply	163.5***	(38.44)	-39.12	(70.71)	-25.32***	(3.746)
10. Financial Services	0.715***	(0.0513)	-1.297***	(0.0932)	-0.0897***	(0.00682)
11. Food Products, Beverages & Tobacco	5.988***	(0.192)	-21.84***	(0.344)	-0.240***	(0.0206)
12. Health and Social Work	1.053	(1.225)	7.454***	(2.197)	-1.698***	(0.149)
13. Hotels and Restaurants	3.842***	(0.174)	-1.028***	(0.340)	0.446***	(0.0181)
14. Machinery, nec	953.7***	(66.81)	696.0***	(124.7)	74.77***	(6.527)
15. Manufacturing, nec; Recycling	26.80***	(5.204)	-37.81***	(9.687)	-1.009*	(0.508)
16. Mining and Quarrying	5.341***	(1.787)	-31.84***	(5.909)	-0.593	(0.528)
17. Other Non-Metallic Mineral Products	4.383***	(0.278)	-19.72***	(0.524)	-1.406***	(0.0429)
18. Other Services	-24.08***	(3.755)	18.53**	(7.276)	1.589***	(0.385)
19. Post and Telecommunication	3.863*	(2.143)	-2.096	(3.958)	-2.345***	(0.307)
20. Public Administration & Defence	0.211	(0.695)	8.478***	(1.481)	-1.145***	(0.0875)
21. Pulp, Paper & Paper Products	-2.328**	(1.042)	-121.6***	(3.532)	-10.06***	(0.380)
22. Rubber and Plastic Products	2.109***	(0.202)	-16.56***	(0.647)	-1.347***	(0.0566)
23. Textiles, Leather & Footwear	3.074***	(0.451)	-2.033	(1.509)	0.153	(0.136)
24. Trade	13.01***	(4.487)	55.50***	(9.727)	0.182	(0.588)
25. Transport Equipment	5.555***	(1.112)	10.35***	(3.026)	-0.451**	(0.218)
26. Transport and Storage	1.534***	(0.347)	0.892	(0.884)	-0.208***	(0.0533)
27. Wood and Products of Wood	-3.544***	(0.836)	-1.389	(2.155)	0.291	(0.233)

Notes: Coefficients from reghdfe specification with Industry FE, Year FE, and Industry-Specific Linear Time Trends ($i.Industry_id * c.Year$). Standard errors in parentheses are clustered at the industry level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Machinery (14) shows anomalously large coefficients across all dimensions, warranting careful interpretation given the heterogeneity within this KLEMS category.

6.2 Service Industries

Business Services (3) shows insignificant employment effects alongside significant productivity gains (13.10^{**}), a clean augmentation signal: the same workforce produces more output. Hotels and Restaurants (13) combines employment growth (3.842^{***}) with quality upgrading (0.446^{***}), suggesting labour-complementary AI in customer-facing roles. Financial Services (10) grows employment (0.715^{***}) while compressing produc-

tivity (-1.297^{***}), consistent with rapid sectoral expansion diluting average output per worker during the fintech-driven hiring surge. The large employment growth coefficient for Electricity, Gas and Water Supply (163.5^{***}) most likely reflects grid digitalisation and renewable capacity expansion rather than AI per se—a limitation of the aggregate KLEMS taxonomy.

6.3 Agriculture

Agriculture (1) shows marginal positive employment effects (0.956^{***}) with insignificant productivity and small quality decline (-0.0995^{**}), consistent with limited AI penetration and predominantly mechanisation-driven change in this sector over the sample period.

7 Discussion

Taken together, the results support a coherent narrative of structured AI-driven workforce restructuring in India rather than uniform job destruction. The static DiD and event study are not contradictory: the positive average EMPG coefficient reflects AI cushioning pre-existing declines in trend-adjusted terms, while the event study reveals that absolute displacement ultimately materialises post-2019, after a five-year adoption lag. The synchronous emergence of employment contraction and quality upgrading after 2019 identifies this as SBTC-consistent restructuring — firms retain and upgrade skilled workers while shedding routine-task workers, consistent with the capital-skill complementarity mechanism of Krusell et al. (2000). These findings refine and extend Behera et al. (2026) in three ways. First, by operating at the industry level rather than the macro aggregate, they reveal the heterogeneity that is obscured in time-series estimation. Second, by restricting the sample to 2009–2022, they avoid conflating AI effects with the 1991 liberalisation and the 2000s IT boom. Third, the Felten task-based index provides a more mechanistically direct measure of AI exposure than patent counts, isolating the occupational channel through which AI reaches workers. Two important limitations should be noted. The crosswalk from Felten’s US-based index to Indian KLEMS industries inherits US occupational structure as a proxy for Indian industry composition — a standard practice, but one that may misclassify exposure in sectors where Indian and US task composition differ substantially. Additionally, the analysis cannot distinguish between formal and informal employment, which is a binding constraint given India’s large informal sector where displacement may be absorbed invisibly.

8 Conclusion

This study provides the first industry-level causal evidence on AI’s labour-market effects in India, using a continuous DiD framework across 27 KLEMS industries from 2009 to 2022. The central findings are: (i) AI exposure raises employment growth on a trend-adjusted basis, but event study evidence reveals a lagged displacement effect materialising post-2019; (ii) labour quality rises concurrently with employment contraction from 2019 onwards, the empirical signature of SBTC; (iii) labour productivity effects are directionally positive but statistically noisy, reflecting heterogeneous firm-level adjustment capacity; and (iv) industry-level effects are highly heterogeneous, with capital-intensive manufacturing restructuring toward high-skill workforces while services exhibit augmentation. The policy implication is direct. India’s demographic dividend — a young, large, predominantly low-skill workforce — intersects with an AI adoption wave that structurally disadvantages workers without cognitive or technical complementarities to AI systems. Reskilling investment at scale, sector-specific transition support, and education-system alignment with AI-complementary skills are not optional long-run considerations but near-term labour market necessities, given that the displacement effects documented here are already statistically detectable in post-2019 data. Future research should extend this analysis to firm-level data to isolate within-industry mechanisms and examine wage and informality effects that this study’s KLEMS framework cannot observe.

8.1 Limitations

Measurement: US Occupational Structure Crosswalk

The Felten index is built on US task data, so mapping it to Indian industries assumes identical occupational compositions across the two economies. Indian manufacturing is more labour-intensive and less technically sophisticated at the baseline, meaning exposure scores may be systematically mismeasured — particularly for labour-intensive sectors — biasing estimates toward zero.

Identification: Single Treatment Threshold

The 2014 Post dummy imposes a common adoption timing across all 27 industries. In reality, fintech, IT services, and manufacturing adopted AI at different speeds. The substantial heterogeneity in Table 4 already signals this — a single threshold averages over very different adoption timelines and may obscure sector-specific dynamics.

Data: Informality

KLEMS captures formal employment only. With roughly 85–90% of India’s workforce in the informal sector, displaced formal workers may simply shift to informal employment rather than exiting the labour market. These displacement estimates are therefore a lower

bound on total labour market disruption.

Data: Absence of Wage Data

The SBTC mechanism you document - fewer but higher-quality workers - should theoretically manifest as a rising skill premium. Without wage data, this channel remains unconfirmed; the LQ result is consistent with SBTC but not conclusive evidence of it.

Aggregation: Within-Industry Heterogeneity

Industry-level coefficients mask firm-level divergence. The positive EMPG estimate for Financial Services, for instance, likely averages over expanding fintech firms and contracting traditional back-office operations simultaneously. Firm-level data would be required to isolate these opposing effects.

References

- [1] Acemoglu, D., & Restrepo, P. (2018). The race between man and machine: Implications of technology for growth, factor shares, and employment. *American Economic Review*, 108(6), 1488-1542.
- [2] Acemoglu, D., & Restrepo, P. (2020). Robots and jobs: Evidence from US labor markets. *Journal of Political Economy*, 128(6), 2188-2244.
- [3] Autor, D. H., Levy, F., & Murnane, R. J. (2003). The skill content of recent technological change: An empirical exploration. *Quarterly Journal of Economics*, 118(4), 1279-1333.
- [4] Behera, D. K., et al. (2026). The debate of augmentation and automation: Does artificial intelligence moderate the growth effectiveness of sectoral employment in India? *Economic Research Forum Working Paper Series*.
- [5] Felten, E., Raj, M., & Seamans, R. (2021). Occupational, industry, and geographic exposure to artificial intelligence: A novel dataset and its potential uses. *Strategic Management Journal*, 44(9), 2195-2217.
- [6] Ganuthula, V. R. R. (2025). Artificial Intelligence and labour market polarisation in India: Strategies for workforce reskilling. *ResearchGate Preprint Archive*.
- [7] IIM Ahmedabad. (2024). Labour-force perception about AI: A study on Indian white-collar workers. *IIMA Research Report Series*.
- [8] Institute of International Finance (IIF). (2026). AI in the Jobs Data: Displacement or Normalization? *Global Disruption Brief Series*.

- [9] Ministry of Electronics & IT. (2026). Labour Market Resilience in Focus as India AI Impact Summit 2026 Discusses Future of Work in AI Era. *Press Information Bureau, Government of India*.
- [10] Press Information Bureau. (2026). India’s Youth Dividend in the AI Era: Accelerating Job Creation and Skilling. *Ministry of Information and Broadcasting, Government of India*.
- [11] Reserve Bank of India. (2023). India KLEMS database: Documentation and user guide. *Department of Economic and Policy Research*.
- [12] Schumpeter, J. A. (1942). *Capitalism, Socialism and Democracy*. New York: Harper & Brothers.
- [13] Upjohn Institute for Employment Research. (2026). AI Exposure and the Future of Work: Tasks, Skill Demand, and Education. *External Papers and Reports*.

A Descriptive Statistics

Table 4 presents the detailed descriptive statistics—specifically the means and standard deviations—for the key KLEMS variables disaggregated across all 27 industries. The data captures the substantial structural heterogeneity of the Indian economy from 2009 to 2022.

Table 4: KLEMS Variables by Industry (Mean & SD)

Industry ID	Emp Growth	LP	LQ	GVA	CS	CC
1	0.396 (4.436)	3.543 (4.835)	0.321 (0.163)	3.939 (2.859)	4.402 (0.876)	0.583 (0.207)
2	3.408 (3.702)	2.563 (9.815)	0.513 (1.009)	5.971 (10.535)	4.814 (4.592)	0.259 (0.459)
3	8.136 (2.972)	1.926 (6.349)	0.530 (0.832)	10.063 (5.849)	9.913 (2.322)	0.984 (0.619)
4	2.918 (4.148)	3.465 (7.235)	1.305 (2.362)	6.384 (5.990)	6.675 (2.120)	0.219 (0.398)
5	3.691 (16.412)	-1.638 (31.822)	0.385 (2.859)	2.054 (27.878)	6.231 (14.667)	-0.232 (0.984)
6	5.109 (3.204)	0.372 (5.759)	0.314 (0.366)	5.481 (5.416)	10.176 (3.035)	0.154 (0.819)
7	2.751 (3.510)	6.288 (4.660)	0.138 (0.397)	9.039 (5.430)	12.447 (1.518)	0.228 (0.390)
8	5.755	0.017	0.238	5.772	5.290	0.416

Continued on next page

Table 4 – continued

Industry ID	Emp Growth	LP	LQ	GVA	CS	CC
	(4.654)	(9.324)	(1.335)	(8.931)	(3.082)	(0.492)
9	3.599	2.339	0.356	5.938	6.178	0.401
	(4.618)	(7.513)	(0.652)	(3.908)	(2.311)	(1.010)
10	3.939	1.705	0.314	5.644	3.077	0.932
	(2.666)	(4.175)	(0.573)	(2.527)	(1.717)	(0.395)
11	0.426	1.794	0.713	2.220	5.750	0.474
	(3.396)	(11.679)	(0.276)	(10.495)	(1.102)	(0.420)
12	4.983	2.300	0.300	7.282	11.542	0.477
	(4.080)	(7.900)	(0.920)	(4.923)	(1.287)	(0.573)
13	3.723	1.653	0.467	5.377	13.320	-0.288
	(2.250)	(25.982)	(0.551)	(26.912)	(4.185)	(0.530)
14	3.036	1.467	-0.015	4.502	7.562	1.429
	(6.336)	(10.489)	(2.920)	(9.469)	(2.354)	(0.619)
15	2.793	8.353	0.308	11.146	3.277	0.883
	(6.244)	(20.059)	(1.059)	(19.853)	(4.359)	(2.517)
16	-2.246	3.537	0.975	1.291	6.231	-0.144
	(6.861)	(11.070)	(2.756)	(8.625)	(5.009)	(0.185)
17	-0.074	5.518	0.859	5.444	4.258	0.694
	(2.832)	(7.746)	(1.075)	(7.088)	(1.390)	(0.468)
18	3.740	0.495	0.662	4.235	4.005	0.636
	(4.990)	(8.176)	(0.607)	(4.590)	(1.349)	(0.258)
19	-0.483	7.499	0.486	7.016	13.239	1.722
	(4.016)	(8.052)	(1.076)	(6.526)	(7.790)	(2.034)
20	0.600	4.538	0.360	5.138	6.714	-0.296
	(2.099)	(5.061)	(0.667)	(4.236)	(0.554)	(0.178)
21	2.245	1.964	-0.206	4.209	0.980	0.589
	(4.389)	(10.796)	(1.502)	(11.101)	(1.280)	(0.356)
22	2.876	5.073	0.475	7.949	7.418	0.709
	(5.086)	(15.713)	(1.858)	(14.716)	(2.834)	(0.456)
23	-0.364	6.156	0.479	5.792	4.816	-0.228
	(3.988)	(10.049)	(0.303)	(9.967)	(2.739)	(0.468)
24	2.852	4.186	0.413	7.038	15.499	0.590
	(4.032)	(9.494)	(0.242)	(8.388)	(7.042)	(0.707)
25	4.091	4.792	0.368	8.882	8.154	0.194
	(4.385)	(15.569)	(1.579)	(13.162)	(3.016)	(0.290)
26	2.458	2.893	0.377	5.351	6.194	1.186
	(1.454)	(10.447)	(0.423)	(10.350)	(2.683)	(1.185)
27	-3.402	8.431	0.399	5.030	2.707	0.563
	(4.584)	(12.635)	(0.277)	(11.535)	(2.630)	(0.339)
Total	2.480	3.379	0.438	5.859	7.069	0.487
	(5.593)	(12.336)	(1.325)	(11.363)	(5.487)	(0.944)

Note: Standard deviations are reported in parentheses below the means.

B Event Study Specification Results

Table 5: Event Study Results

Variables	Employment Growth	Labor Productivity	Labor Quality
AI_x_2010	-0.194 (0.274)	-1.295 (2.295)	0.098* (0.057)
AI_x_2011	-0.416 (0.449)	0.359 (3.935)	0.193* (0.117)
AI_x_2012	0.205 (0.272)	1.191 (2.245)	-0.004 (0.028)
AI_x_2014	-0.118 (0.426)	-1.026 (4.781)	0.091 (0.068)
AI_x_2015	-0.461 (0.515)	-1.129 (3.804)	0.188 (0.120)
AI_x_2016	-0.731 (0.720)	-1.538 (4.185)	0.287 (0.181)
AI_x_2017	-0.990 (0.923)	2.114 (4.396)	0.384 (0.241)
AI_x_2018	-0.983 (2.077)	-0.752 (6.281)	0.557 (0.478)
AI_x_2019	-3.571* (2.115)	6.330 (5.625)	0.429 (0.723)
AI_x_2020	-4.976** (2.204)	7.701 (7.875)	1.199** (0.500)
AI_x_2021	-4.430* (2.471)	3.706 (7.611)	1.309* (0.723)
AI_x_2022	-4.227 (2.860)	5.940 (8.800)	1.554* (0.882)
CS	0.186* (0.097)	0.130 (0.212)	-0.013 (0.018)
CC	-0.191 (0.641)	0.863 (1.063)	0.030 (0.079)
Observations	378	378	378
R-squared	0.356	0.207	0.239

Notes: Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. All models include Industry FE, Year FE, and Industry-Specific Linear Time Trends. Base year: 2013.